Spatial Data Analysis

AGRON 5130 - Module 12 Assignment

Caio dos Santos

Introduction

The goal of this assignment is to practice the full workflow from Module 12:

- loading and inspecting spatial vector data
- checking and using coordinate reference systems
- joining spatial datasets
- interpreting scale and resolution effects
- building interpolated surfaces (rasters) with kriging
- converting predictions into management recommendations

You will complete two parts:

- **Part 1:** vector data exploration and spatial joins
- **Part 2:** raster creation and spatial pattern analysis

Submit a knitted HTML/PDF generated from your R Markdown file to Canvas.

Setup

```
library(sf)
```

```
## Linking to GEOS 3.12.1, GDAL 3.8.4, PROJ 9.4.0; sf_use_s2() is TRUE
```

```
library(gstat)
library(stars)
```

```
## Loading required package: abind
```

Part 1: Vector Data and Spatial Joins

Question 1

2 points

Read in the SSURGO polygon layer (`data/sample-ssurgo.shp`) and the soil test data (`data/soil_tests.shp`).

For each dataset, report:

1. the geometry type
2. the CRS
3. whether the CRS is longlat or projected

Interpretation: If the CRS is longlat, which one should you transform to a projected system, and why?

Question 2

2 points

Transform both datasets to the same projected CRS (use 32615 for UTM zone 15N).

Make a map showing both layers together. Use `plot()` to overlay SSURGO polygons (border only) with soil test points on top.

Question 3

2 points

Use `st_join()` with `st_within` to assign SSURGO attributes (for example `mukey`, `muname`, `om_r`) to each soil test point.

Question 4

2 points

Compute the mean `measure` value by SSURGO map unit and attribute (use `aggregate()`).

Include `mukey` in the grouping variables.

Part 2: Raster Creation and Spatial Pattern

Question 5

1 point

Load the field boundary (`data/boundary.shp`) and reproject (using `st_transform`) it to EPSG 32615. Create a raster grid that covers the boundary and has a 10 m resolution using `st_as_stars()` and `st_crop()`.

Plot your empty grid.

Question 6

2 points

Choose the soil attribute `P_bray` from the original soil test points (not the boundary). Compute the empirical variogram with `variogram()` and plot it.

Question 7

2 points

Using your selected attribute from Question 6, fit a spherical model (`"Sph"`) with `fit.variogram()` and plot empirical + fitted variogram together.

Fit a spherical model (`"Sph"`) with `fit.variogram()`, and plot empirical + fitted variogram together.

Question 8

2 points

Using the fitted model from Question 7, run `krige()` across the grid and map the predicted surface (`var1.pred`).

Run `krige()` across the grid and map the predicted surface (`var1.pred`).